

Phase 4: Initial System Configuration & Security Implementation (Week 4):

1. Configure SSH with key-based authentication

Step performed (via SSH)

1. on the workstation:

```
ramesh@rameshserver:~$ ssh-keygen
Generating public/private ed25519 key pair.
```

Copy the public key to the server:

```
ramesh@rameshserver:~$ ssh-copy-id ramesh@192.168.56.101
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/ramesh/.ssh/id_ed25519.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
ramesh@192.168.56.101's password:

Number of key(s) added: 1

Now try logging into the machine, with:  "ssh 'ramesh@192.168.56.101'"
and check to make sure that only the key(s) you wanted were added.
```

Test SSH key login:

```
ramesh@rameshserver:~$ ssh ramesh@192.168.56.101
Enter passphrase for key '/home/ramesh/.ssh/id_ed25519':
ramesh@192.168.56.101's password:
Permission denied, please try again.
ramesh@192.168.56.101's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-90-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue 23 Dec 11:36:34 UTC 2025

System load:          0.0
Usage of /:            31.2% of 23.49GB
Memory usage:         6%
Swap usage:           0%
Processes:            133
Users logged in:      1
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fec4:9ac

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Mon Dec 22 00:07:51 2025 from 192.168.56.101
```

Configuration Change

File edited:

```
# This is the sshd server system-wide configuration file. See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

# When systemd socket activation is used (the default), the socket
# configuration must be re-generated after changing Port, AddressFamily, or
# ListenAddress.
#
# For changes to take effect, run:
#
#   systemctl daemon-reload
#   systemctl restart ssh.socket
#
#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
#PermitRootLogin prohibit-password
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10
```

[Read 131 lines]

Restart SSH (Ubuntu 24.04):

```

ramesh@rameshserver:~$ sudo systemctl restart ssh
ramesh@rameshserver:~$ sudo systemctl status ssh.socket
• ssh.socket - OpenBSD Secure Shell server socket
  Loaded: loaded (/usr/lib/systemd/system/ssh.socket; enabled; preset: enabled)
  Active: active (running) since Tue 2025-12-23 13:04:36 UTC; 24min ago
  Triggers: • ssh.service
  Listen: 0.0.0.0:22 (Stream)
           [::]:22 (Stream)
  Tasks: 0 (limit: 4602)
  Memory: 8.0K (peak: 260.0K)
  CPU: 3ms
  CGroup: /system.slice/ssh.socket

Dec 23 13:04:36 rameshserver systemd[1]: Listening on ssh.socket - OpenBSD Secure Shell server socket.

```

Test SSH login:

```

Dec 23 13:04:36 rameshserver systemd[1]: Listening on ssh.
ramesh@rameshserver:~$ ssh ramesh@192.168.56.101
Enter passphrase for key '/home/ramesh/.ssh/id_ed25519':
Enter passphrase for key '/home/ramesh/.ssh/id_ed25519':
Enter passphrase for key '/home/ramesh/.ssh/id_ed25519':
ramesh@192.168.56.101's password:
Connection closed by 192.168.56.101 port 22

```

2. configure a firewall permitting SSH from one specific workstation only?

1.Firewall Configuration (UFW)

Enable firewall:

```

All packages are up to date.
ramesh@rameshserver:~$ sudo ufw enable
Firewall is active and enabled on system startup
ramesh@rameshserver:~$ sudo ufw allow from 192.168.56.1 to any port 22
Rule added
ramesh@rameshserver:~$ sudo ufw deny ssh
Rule added
Rule added (v6)

```

3.Manage users and implement privilege management, creating a non-root administrative user.

```
ramesh@rameshserver:~$ sudo adduser adminuser
fatal: The user `adminuser' already exists.
ramesh@rameshserver:~$ sudo usermod -aG sudo adminuser
ramesh@rameshserver:~$ groups adminuser
adminuser : adminuser sudo users
```

4.SSH Access Evidence showing successful connection screenshots?

Ans: **SSH Access Evidence**

Evidence Required

Screenshot showing:

```
ramesh@rameshserver:~$ ssh adminuser@192.168.56.101
adminuser@192.168.56.101's password:
Permission denied, please try again.
adminuser@192.168.56.101's password:
Permission denied, please try again.
adminuser@192.168.56.101's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-90-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue 23 Dec 13:54:11 UTC 2025

System load:            0.02
Usage of /:              31.5% of 23.49GB
Memory usage:           6%
Swap usage:             0%
Processes:              139
Users logged in:        1
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fec4:9ac

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The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
```

6. Firewall Documentation showing complete ruleset:

```
adminuser@rameshserver:~$ sudo ufw status verbose
[sudo] password for adminuser:
Sorry, try again.
[sudo] password for adminuser:
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To Action From
--
22 ALLOW IN 192.168.56.1
22/tcp DENY IN Anywhere
22/tcp (v6) DENY IN Anywhere (v6)
```

7. Remote Administration Evidence demonstrating commands executed via SSH?

```
# possible, but leave them commented. Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

# When systemd socket activation is used (the default), the socket
# configuration must be re-generated after changing Port, AddressFamily, or
# ListenAddress.
#
# For changes to take effect, run:
#
#   systemctl daemon-reload
#   systemctl restart ssh.socket
#
#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
#PermitRootLogin prohibit-password
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10
```

```
adminuser@rameshserver:~$ sudo ufw status
Status: active
```

To	Action	From
--	-----	----
22	ALLOW	192.168.56.1
22/tcp	DENY	Anywhere
22/tcp (v6)	DENY	Anywhere (v6)